Zaú Júlio

Tech Lead | Full Stack Software Engineer | Microservices & System Design | AI / ML

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PROFESSIONAL PROFILE

Tech Lead and hands-on Full Stack Software Engineer with 5+ years building scalable, high-concurrency systems across the full software development life cycle. Deep expertise in the **TypeScript / Node.js** ecosystem (**GraphQL**, Express, NestJS) and in **Python** for **AI, Machine Learning, Deep Learning, NLP and time-series** analytics. Designs event-driven **microservices** backed by job queues (BullMQ, RabbitMQ), models relational and NoSQL data (PostgreSQL, MySQL/MariaDB, MongoDB), and owns delivery end-to-end—from ER modeling and **C# / .NET** services to CI/CD, Linux infrastructure and cloud cost optimization. Ranked among the **top worldwide in GraphQL on CodersRank**. Started engineering before the AI wave and is now fluent at AI-augmented development, leading teams while still shipping production code daily.

TECHNICAL SKILLS

- **Programming Languages:** TypeScript / JavaScript, Python, C#, Java, C, Bash, SQL.
- **Backend & Microservices:** Node.js, Express, NestJS, GraphQL, .NET / .NET Core, FastAPI, Django, REST, WebSockets / Socket.io, BullMQ, RabbitMQ.
- **Frontend:** React, Vite, Next.js, Redux, Tailwind CSS, Styled Components, HTML / CSS, SSR, Core Web Vitals.
- **Databases:** PostgreSQL, MySQL / MariaDB, MongoDB, Redis, Data Lakes, ER modeling.
- **AI / Data:** Machine Learning, Deep Learning, NLP, Time Series, Pandas, NumPy, scikit-learn, PyTorch, Matplotlib, Jupyter, R.
- **Infra / DevOps:** Docker, Linux, Shell scripting, CI/CD, AWS, GCP, VPC, CDN, DevSecOps, Monitoring / Observability.
- **Practices:** System design, microservices architecture, SDLC, code review, technical leadership, AI-augmented development (Claude, OpenCode).

EXPERIENCE

Bitwise Technology — Sorocaba, SP (Remote)

Tech Lead — Aug 2025 – Present

- Define the engineering roadmap and technical standards for TypeScript microservices and GraphQL schemas, owning system design end-to-end (ER modeling to deployment) while staying hands-on and shipping production code daily.
- Drove targeted infrastructure optimization that cut CI/CD build times by 75% (20 min to under 5) and lowered monthly AWS spend by ~80% ($120 to $25), while right-sizing Node.js production services to half the compute and memory footprint with no impact on throughput.
- Mentor cross-functional engineers and lead code reviews, instilling a culture of quality, scalability and observability.

Full Stack Software Engineer — Mar 2023 – Aug 2025

- Delivered high-availability backends in GraphQL, Node.js, C#/.NET and MongoDB, modeling complex domain logic and offloading heavy processing to BullMQ and RabbitMQ job queues.
- Built fast, SEO-friendly React/Vite/Next.js frontends tuned for SSR and Core Web Vitals, plus the CI/CD pipelines and cloud provisioning that shortened release cycles.

Commcepta **Dec 2021 – Dec 2022**
Full Stack Developer Curitiba, PR (Remote)

- Tuned GraphQL queries over a large-scale Data Lake, cutting search latency from ~30s to under 6s (up to 50% faster) and enabling near real-time analytics on Big Data.
- Engineered Python pipelines that unified multiple data sources for enterprise decision-making, and hardened a multinational portal for accessibility (a11y) and internationalization (i18n).

Byte Seridó Jr – Junior Enterprise (UFRN) **Jun 2021 – Nov 2022**
Project Director / Front-end Developer Caicó, RN

- Led the planning and delivery of multiple software projects at the university's junior enterprise, progressing from front-end developer to Project Director.
- Mentored and tutored incoming developers, onboarding students on programming fundamentals, engineering best practices and project workflows.

RESEARCH & PUBLICATION

LabEPI / UFRN **Aug 2019 – Sep 2020**
AI & Machine Learning Researcher (Scientific Initiation) Caicó, RN

- Designed a hybrid classifier combining **Self-Organizing Maps (SOM)** neural networks with Polynomial Regression for unsupervised anomaly detection in minute-level energy data, improving accuracy over a pure K-Means baseline and reducing false positives.
- Optimized AI model generation and reuse, cutting model memory footprint from 2 GB to ~200 MB and SOM training time from 10–15 minutes to under 2 minutes.
- Authored a peer-reviewed paper published at the Congress of Scientific and Technological Initiation (CICT); project PVF16985-2019, "Intelligent Management of Energy Consumption" (GERINCE).

SELECTED OPEN-SOURCE PROJECTS

FeaturesAnalyzer [Python] — Time-series analysis and machine-learning toolkit for feature exploration, modeling and forecasting over high-volume datasets.

ZSOM [Python] — From-scratch implementation of the Self-Organizing Maps algorithm, powering the unsupervised models from the energy-consumption research.

weasyprint-pdf-render [Python, Docker] — High-performance HTML-to-PDF rendering microservice, containerized for horizontal scaling.

linuwu-sense-dkms [C, Shell] — DKMS kernel package for the Linuwu-Sense driver (Acer WMI Extras), demonstrating low-level Linux systems work.

EDUCATION

UFRN – Federal University of Rio Grande do Norte 2019 – 2025
B.Sc. in Information Systems — honors thesis (TCC) graded 10/10

IMD/UFRN – Digital Metropolis Institute 2018 – 2019
Technical Degree in Computer Networks

LANGUAGES & AWARDS

- **CodersRank:** #1 worldwide in GraphQL (currently #2); #1 in João Pessoa and Top 150 in Brazil overall.
- **Awards:** 2nd place — 1st UFRN Programming Marathon (regional campus); Research Grant PVF16985-2019 (GERINCE).
- **Spoken Languages:** Portuguese (Native), English (Professional Working Proficiency), Spanish (Working Proficiency – Listening).